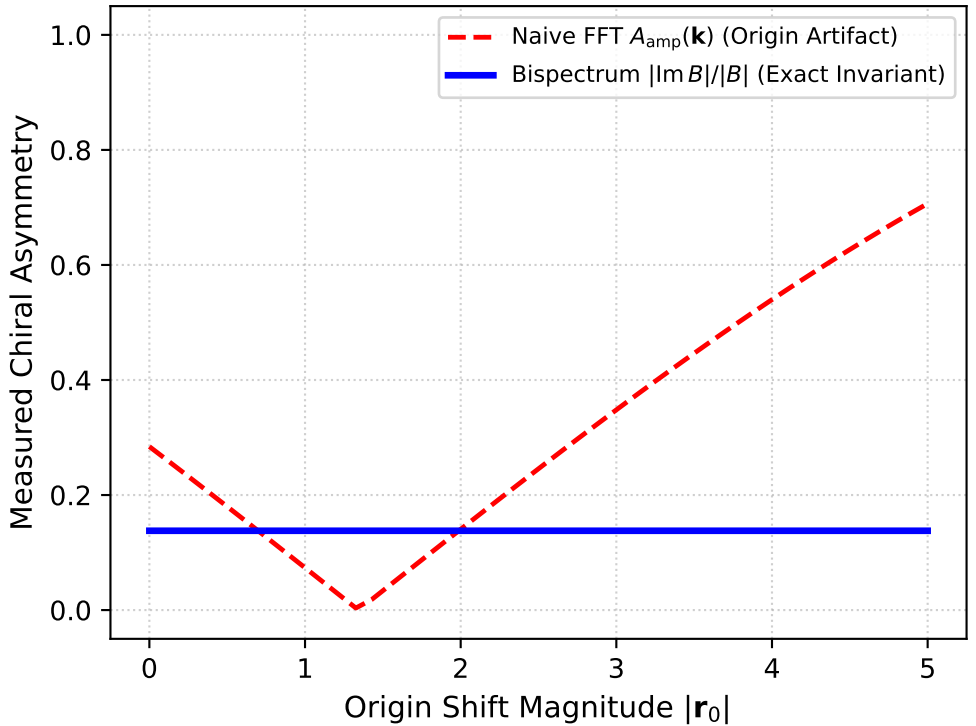
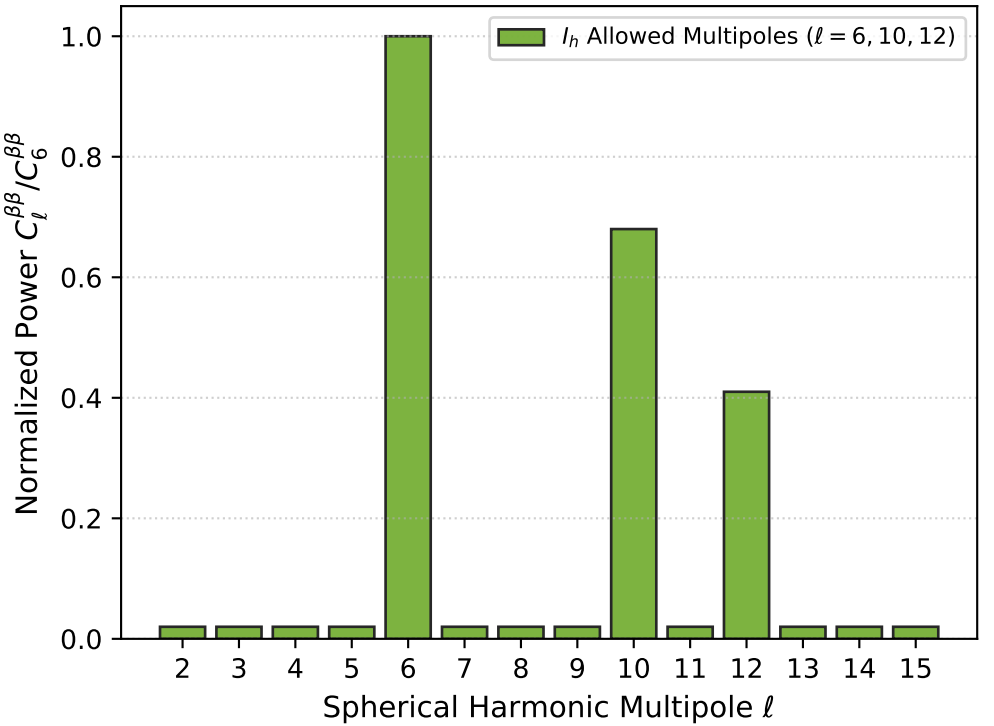

(a) Origin-Shift Invariance Audit ($\mathbf{r} \to \mathbf{r} + \mathbf{r}_0$)
Naive FFT $A_{\text{amp}}(\mathbf{k})$ (Origin Artifact)
Bispectrum $|\text{Im}\, B|/|B|$ (Exact Invariant)
Measured Chiral Asymmetry
Origin Shift Magnitude $|\mathbf{r}_0|$
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5


(b) CMB Anisotropic Birefringence $C_\ell^{\beta\beta}$ Locking
I_h Allowed Multipoles ($\ell = 6, 10, 12$)
Normalized Power $C_\ell^{\beta\beta}/C_6^{\beta\beta}$
Spherical Harmonic Multipole ℓ
0.0
0.2
0.4
0.6
0.8
1.0
2
3
4
5
6
7
8
9
10
11
12
13
14
15